

SYNOPSIS OF THE PROPOSED RESEARCH PROJECT
FOR THE NATIONAL SCIENCE AND TECHNOLOGY (NST) FELLOWSHIP, 2025-26

Prevalence and predictors of food insecurity among 6-23 month-old children in Bangladesh

1. Introduction

Food insecurity represents a critical threat to child survival, growth, and development, particularly during the complementary feeding period when children aged 6-23 months are most vulnerable to nutritional deficiencies [1]. Food insecurity among children aged 6-23 months in Bangladesh represents a critical public health concern that demands urgent research attention [2]. The 6-23 month age group constitutes a critical window for child development; where proper nutrition is fundamental for growth, cognitive development, and long-term health outcomes [2]. Research demonstrates that 36.3% of children aged 6-23 months in Bangladesh experience household food insecurity [2], while two-thirds of children under five face child food poverty [3]. This alarming prevalence, combined with 23.6% of children under five being stunted and 11.0% being wasted [3], establishes the urgent need for comprehensive research on food insecurity predictors in this vulnerable population [4].

The topic is aligned with Bangladesh's national nutrition policies, which emphasize the first 1,000 days of life as a critical intervention period [5, 6], which specifically target the first 1000 days of life as a priority intervention period. It also supports SDG 2.2, aiming to end malnutrition among children under five by 2030 [7, 8]. Despite the recognized burden of food insecurity, limited research exists that focuses specifically on the 6-23-month age group using validated tools. The Food Insecurity Experience Scale (FIES) has already been validated for use in rural Bangladesh [9], and the recent development of the Early Childhood FIES (EC-FIES) by UNICEF provides an even more targeted approach to measuring food insecurity among young children [10]. By investigating both the prevalence and the predictors of food insecurity among children aged 6-23 months in Bangladesh, this study seeks to fill critical evidence gaps and support the design of effective, context-specific interventions.

2. Objective of the study

2.1 General objective

To assess the prevalence of food insecurity and identify its key sociodemographic and health-related predictors among children aged 6-23 months in Bangladesh, with the aim of informing targeted interventions to improve child nutrition and well-being.

2.2 Specific objectives

- To estimate the national and regional prevalence of food insecurity among children aged 6-23 months in Bangladesh
- To examine the characteristics of households with food-insecure children
- To identify key predictors of food insecurity among children aged 6-23 months
- To assess the association between food insecurity and child feeding practices
- To recommend context-specific interventions and policies

3. Methodology

This rapid assessment will apply a cross-sectional study design to collect quantitative data on the diets of children aged 6-23 months and the barriers experienced by the mothers in feeding nutritious diets to these children.

3.1 Study area and population

Data will be collected from 25 upazilas from 8 randomly selected districts of 8 divisions. Quantitative data on sociodemographic characteristics, household income and expenditure, minimum dietary diversity and barriers to feeding children will be collected from the mothers of children aged 6-23 months.

3.2 Sample size calculation

Health related congregation sites of upazilas will serve as primary sampling unit which will be drawn by a stratified multistage random sampling. From these congregation sites, mothers of 6-23 months old children will be chosen by convenience sampling.

Table 1. Design matrix

Study area	Division	Total Upazila	Interview with mothers
1	Dhaka	4	64
2	Sylhet	3	48
3	Rajshahi	3	48
4	Mymensingh	3	48
5	Chattogram	3	48
6	Khulna	3	48
7	Barishal	3	48
8	Rangpur	3	48
Total		25	400

3.3 Data collection

3.3.1 Childhood Food Insecurity Data: Childhood food insecurity will be assessed using the Early Childhood Food Insecurity Experience Scale (EC-FIES), developed by UNICEF in 2025 to measure food access challenges among 6–23-month-old children. The tool includes eight standardized questions and is validated for use in low- and middle-income countries like Bangladesh.

3.3.2 Sociodemographic data: Participants will be administered a semi-structured questionnaire comprising a series of questions to gather sociodemographic data, including age, gender, education level, income, and other relevant information.

3.3.3 Complementary Feeding Practices: Assessment of complementary feeding practices will follow WHO/UNICEF indicators: Minimum Dietary Diversity (MDD), Minimum Meal Frequency (MMF), Minimum Acceptable Diet (MAD), and Consumption of iron-rich foods.

3.4 Data analysis

Different types of descriptive and inferential statistics will be performed with the quantitative data using SPSS version 26. Parametric tests will be performed for normal data and for non-normal data nonparametric tests will be performed.

3.5 Ethical approval

All the required ethical approvals will be obtained from the appropriate ethics review board and all participants will provide informed consent before participating in the study.

4. Operational planning

Activities	2024-25											
	Months											
	1	2	3	4	5	6	7	8	9	10	11	12
Inception Preparation	■											
Literature Review		■										
Methodology Preparation			■	■								
Enumerator Recruitment					■							
Enumerator Training						■						
Data Collection						■	■	■				
Data Cleaning & Analysis								■	■	■		
Thesis Writing										■	■	■
Dissemination												■

5. Socioeconomic importance of the proposed research

This study addresses critical knowledge gaps in understanding food insecurity among the most vulnerable population children aged 6-23 months in Bangladesh. Results will inform evidence-based policy interventions targeting child food insecurity, supporting Bangladesh's commitment to achieving Sustainable Development Goals 2 (Zero Hunger) and 3 (Good Health and Well-being). This study will be among the first to validate and apply the newly developed EC-FIES tool in the Bangladeshi context, contributing to the global evidence base for food insecurity measurement tools. By identifying barriers to food security and understanding the magnitude of the problem, the study will contribute to improving child health outcomes and reducing malnutrition in Bangladesh.

6. References

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